

ADARSH SAURABH

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Education

National Institute Of Technology, Rourkela <i>Master of Technology in Signal and Image Processing, CGPA: 8.28</i>	2025 – 2027 <i>Rourkela, India</i>
Guru Ghasidas University, Bilaspur <i>Bachelor of Technology in Computer Science and Engineering, CGPA: 8.50</i>	2020 – 2024 <i>Bilaspur, India</i>
Jawahar Navodaya Vidyalaya, JNV Sitamarhi <i>Matriculation CBSE, CGPA: 8.8</i>	2017 <i>Sitamarhi, India</i>

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Artificial Intelligence
- Computer Vision
- Software Methodology
- Database Management
- Computer Architecture
- Soft Computing

Experience

Autobot Robotics <i>Machine Learning Intern</i>	May 2023 – Aug 2023 <i>Remote</i>
• Mentored and guided a junior intern, imparting mastery in ML fundamentals, Data Manipulation, NumPy, and Pandas. • Developed and deployed (Django) 4 ML/DL projects with models for integration into company robotic systems.	
Ziroh Labs Source Link <i>Team Leader - Academic Alliance Program</i>	Jan 2022 – Feb 2022 <i>Remote</i>
• Fostered strong mentor-team connections, facilitating seamless communication and collaboration. • Successfully led a cross-functional team of 5 to achieve image processing project completion ahead of schedule.	

Key Projects

Amazon ML Challenge 2026 Source Link <i>Python, Polars, CatBoost GPU, LightGBM, RapidFuzz</i>	2026
• Engineered end-to-end production entity resolution pipeline over 24.2M records using Polars lazy execution and multi-view blocking. • Built 14-dim C++ RapidFuzz feature extraction engine; trained CatBoost GPU and LightGBM with Isotonic calibration achieving Macro F0.5 = 0.9719.	
Alternative Data Radar Source Link <i>Python, Next.js, SQL, Bright Data, LLM APIs</i>	2026
• Engineered automated data collection backend in Next.js to gather public web signals (careers and pricing data), routing requests via Bright Data Web Unlocker to bypass anti-scraping mechanisms. • Implemented AI-driven analysis to calculate a 0–100 corporate health score stored in a SQL database, presenting pre-earnings intelligence and warning alerts in a responsive Recharts dashboard.	
Apna Gold Solutions — Multi-Tenant SaaS Platform Source Link <i>Django REST API, React Vite, JWT</i>	2025
• Developing multi-tenant B2B2C platform with seller registration, KYC verification, and customer subscription plans. • Implemented custom JWT authentication with database conflict resolution and secure tenant data isolation. • Built notification systems, OTP verification, and real-time status tracking for complex business workflows.	
Uplan — Adversarial Document Intelligence Pipeline Source Link <i>Python, Gemini API, LangGraph, Streamlit</i>	2025
• Architected adversarial multi-agent workflow in LangGraph utilizing Gemini 2.5 Pro specialists to check visa document coherence; reduced manual verification workload by 85% with detailed failure rebuttal generation. • Designed structural encoding system using Gemini 2.0 Flash to compile page-level document metadata into a typed semantic graph (98% token compression) and integrated zero-hallucination mathematical rule check engine.	

CodeAtlas — AST Architecture Analyzer Source Link <i>Python, AST, Graph Algorithms, Mermaid.js</i>	2026
- Engineered zero-dependency static analysis engine using standard library <code>ast</code> to parse function hierarchies, imports, and cross-module dependencies without executing user code. - Implemented automated anti-pattern detection for nested loops and $N + 1$ database queries, rendering interactive codebase architecture graphs via Mermaid.js in a local web workbench.	
IBYD Technology — Warehouse PathMapper Source Link <i>Python, Algorithms, Data Analysis</i>	2023
- Delivered dynamic warehouse mapping solution as a freelance engagement, working directly with client to translate operational pathfinding needs into a heuristic-based routing system. - Scaled routing engine to warehouses as large as 10,000×10,000 units, computing optimal paths through 10,000+ locations simultaneously in under 0.5 seconds on a standard i5 laptop CPU.	
K-HOG Unsupervised Keyframe Identifier (K-HUKI) Source Link <i>Python, Computer Vision, ML</i>	2023
- Built keyframe extraction system with HOG Features and Unsupervised Learning achieving 96.45% accuracy. - Developed preprocessing architecture 11 times faster than ResNet-based approaches with real-time processing.	

Achievements

Intermediate Winner — Awaiken Hacks 2023 Source Link : Innovative tool using MediaPipe for real-time action analysis in education with precise feedback on posture and motion.
Rank 1 — Girlsript Bilaspur Hackathon "Hack4CG" Source Link : Won top honors for Facial Emotion Recognition project addressing depression vulnerability in modern society.

Leadership / Extracurricular

Google Developers Student Club <i>Vice President</i>	3rd Year <i>Guru Ghasidas University</i>
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Technical Skills

Languages: Python, C, C++, HTML, CSS, SQL, TypeScript Developer Tools: VS Code, Git, GitHub, CUDA, Anaconda Technologies/Frameworks: Polars, CatBoost (GPU), LightGBM, RapidFuzz, Django, Flask, Next.js, React Vite, LangGraph, Streamlit, NumPy, Pandas, TensorFlow, Scikit-learn, OpenCV, Supabase, Docker Specializations: Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, LLMOps, Computer Vision, Multi-Agent Systems, Data Structures & Algorithms
